Symmetric tensor stability implies tensor stability

Candidate full solution for arXiv:1603.00246

11 August 2026

Status. Candidate full proof, likely valid, requiring human review. The answer to the source question is affirmative. In fact, stability under symmetric squares alone implies full tensor stability.

1 Source question

Let $0 < p \leq 1$ and let $(\mathcal{I}, \|\cdot\|_{\mathcal{I}})$ be a p-normed operator ideal over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Botelho and Torres call $\mathcal{I}$ *symmetrically tensorstable* (or *s-tensorstable*) if, for every $m \in \mathbb{N}$, there is $C_m > 0$ such that

$$\otimes^{m,s} u \in \mathcal{I}\big(\widehat{\otimes}_{\pi_s}^{m,s} E; \widehat{\otimes}_{\pi_s}^{m,s} F\big), \qquad \| \otimes^{m,s} u\|_{\mathcal{I}} \leq C_m \|u\|_{\mathcal{I}}^m$$

for all Banach spaces E, F and $u \in \mathcal{I}(E; F)$.

They call $\mathcal{I}$ *tensorstable* if there are constants $D_m > 0$ such that for every $u_j \in \mathcal{I}(E_j; F_j)$,

$$u_1 \otimes \cdots \otimes u_m \in \mathcal{I}(E_1 \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi E_m; F_1 \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi F_m)$$

and

$$\|u_1 \otimes \cdots \otimes u_m\|_{\mathcal{I}} \leq D_m \prod_{j=1}^m \|u_j\|_{\mathcal{I}}.$$

After proving the converse implication, they ask at the end of the paper:

> "We do not know if every s-tensorstable operator ideal is tensorstable."

This is the open question on PDF page 27 (printed page 26) of [1].

Open question. We do not know if every s-tensorstable operator ideal is tensorstable. If yes, Example 7.9 will have a simpler reasoning (in [31, Corollary 34.3.1] it is proved that Π_p is not tensorstable); and Theorem 7.8 will have one more equivalent condition. But the reader should have in mind that things that work well for homogeneous polynomials might not work so well for multilinear operators. For example, Leung [46] proved that the dual J' of the James space J is symmetrically regular but fails to be regular.

Figure 1: The complete open question in [1, p. 27], rendered and cropped from the local source-paper PDF.

2 Proof intuition

Put two ideal operators on the diagonal of one operator between finite ℓ_1-sums. The symmetric square of this diagonal operator has three coordinate types: the square of the first block, the square of the second block, and the mixed block. That mixed block is exactly the ordinary projective tensor product of the two original operators.

There are uniformly bounded maps which insert an ordinary tensor into this mixed symmetric block and extract it again. Thus symmetric-square stability of the diagonal operator forces the ordinary tensor product to belong to the ideal. Normalizing the two diagonal blocks before adding them converts the p-triangle estimate into the required product bound. Once the two-operator estimate is available, associativity gives arbitrary finite tensor products by iteration.

3 The mixed component of a symmetric square

For a Banach space X, write

$$z \vee z' = \frac{1}{2}(z \otimes z' + z' \otimes z) \in \widehat{\otimes}^{2,s}_{\pi_s} X.$$

This is the symmetrization convention used in [1, Proposition 7.12].

Lemma 1. *Let E, G, F, H be Banach spaces. Put $X = E \oplus_1 G$ and $Y = F \oplus_1 H$, with canonical coordinate injections i_E, i_G, i_F, i_H and projections q_E, q_G, q_F, q_H. There are bounded operators*

$$J : E\widehat{\otimes}_\pi G \longrightarrow \widehat{\otimes}^{2,s}_{\pi_s} X, \qquad Q : \widehat{\otimes}^{2,s}_{\pi_s} Y \longrightarrow F\widehat{\otimes}_\pi H$$

such that

$$J(x \otimes g) = i_E(x) \vee i_G(g), \qquad Q(i_F(f) \vee i_H(h)) = f \otimes h,$$

and $\|J\| \le 2$, $\|Q\| \le 2$.

Proof. Let $S^2_X : X \times X \to \widehat{\otimes}^{2,s}_{\pi_s} X$ be the symmetrization map $S^2_X(z, z') = z \vee z'$. Its linearization on $X\widehat{\otimes}_\pi X$ has norm equal to the second polarization constant, which is at most 2; this is precisely the estimate used in [1, Proposition 7.12]. Therefore

$$J = (S^2_X)_L \circ (i_E \otimes i_G)$$

has norm at most 2 and has the required action on elementary tensors.

For completeness, the same bound follows directly without identifying any two tensor norms. The polarization identity

$$z \vee z' = \frac{1}{4}\big((z + z')^{\otimes 2} - (z - z')^{\otimes 2}\big)$$

and replacement of (z, z') by $(tz, t^{-1}z')$, followed by minimization over $t > 0$, give

$$\pi_s(z \vee z') \le 2\|z\|\,\|z'\|.$$

The universal property of $\widehat{\otimes}_\pi$ then yields the asserted extension of J.

Let

$$\iota^2_Y : \widehat{\otimes}^{2,s}_{\pi_s} Y \longrightarrow Y\widehat{\otimes}_\pi Y$$

be the canonical norm-one inclusion used in [1, Proposition 7.12], and define

$$Q = 2(q_F \otimes q_H) \circ \iota^2_Y.$$

Then $\|Q\| \le 2$. Moreover,

$$\iota_Y^2(i_F(f) \vee i_H(h)) = \frac{1}{2}\big(i_F(f) \otimes i_H(h) + i_H(h) \otimes i_F(f)\big).$$

The coordinate tensor projection $q_F \otimes q_H$ keeps the first term and annihilates the second, so the factor 2 gives $Q(i_F(f) \vee i_H(h)) = f \otimes h$. $\square$

4 Affirmative solution

Theorem 1. *Let $0 < p \le 1$. Every s-tensorstable p-normed operator ideal is tensorstable. More precisely, suppose only that symmetric squares satisfy*

$$\otimes^{2,s} w \in \mathcal{I}(\widehat{\otimes}_{\pi_s}^{2,s} X; \widehat{\otimes}_{\pi_s}^{2,s} Y), \qquad \| \otimes^{2,s} w\|_{\mathcal{I}} \le C_2 \|w\|_{\mathcal{I}}^2$$

for all $w \in \mathcal{I}(X;Y)$. Then, for all $u \in \mathcal{I}(E;F)$ and $v \in \mathcal{I}(G;H)$,

$$\|u \otimes v\|_{\mathcal{I}} \le K \|u\|_{\mathcal{I}} \|v\|_{\mathcal{I}}, \qquad K = 2^{,2+2/p} C_2. \tag{1}$$

Consequently one may take $D_m = K^{m-1}$ in the definition of tensorstability.

Proof. The conclusion is immediate if $u = 0$ or $v = 0$, so suppose both are nonzero and set

$$a = \|u\|_{\mathcal{I}}, \qquad b = \|v\|_{\mathcal{I}}.$$

Use the spaces and coordinate maps of Lemma 1 and define the block-diagonal operator

$$W = i_F \frac{u}{a} q_E + i_H \frac{v}{b} q_G : X \longrightarrow Y.$$

The ideal property puts both summands in $\mathcal{I}(X;Y)$ with ideal p-norm at most one. Since $\mathcal{I}$ is p-normed,

$$\|W\|_{\mathcal{I}}^p \le 1 + 1 = 2, \qquad \text{hence} \qquad \|W\|_{\mathcal{I}}^2 \le 2^{2/p}. \tag{2}$$

The defining formula for $\otimes^{2,s} W$ on pure squares, polarized, gives

$$(\otimes^{2,s} W)(z \vee z') = Wz \vee Wz' \qquad (z, z' \in X). \tag{3}$$

Thus, for $x \in E$ and $g \in G$, Lemma 1 and the diagonal form of W yield

$$\begin{aligned} Q(\otimes^{2,s} W) J(x \otimes g) &= Q\left(i_F \frac{u(x)}{a} \vee i_H \frac{v(g)}{b} \right) \\ &= \frac{u(x)}{a} \otimes \frac{v(g)}{b}. \end{aligned}$$

By density of elementary tensors,

$$Q \circ (\otimes^{2,s} W) \circ J = \frac{u}{a} \otimes \frac{v}{b}. \tag{4}$$

Symmetric-square stability, the operator-ideal property, Lemma 1, and (2) now give

$$\begin{aligned} \Big\| \frac{u}{a} \otimes \frac{v}{b} \Big\|_{\mathcal{I}} &\le \|Q\| \, \| \otimes^{2,s} W\|_{\mathcal{I}} \, \|J\| \\ &\le 2\, C_2 \|W\|_{\mathcal{I}}^2 \, 2 \\ &\le 2^{,2+2/p} C_2 = K. \end{aligned}$$

Multiplication by ab proves (1).

Finally apply (1) repeatedly. Identifying iterated projective tensor products through their canonical isometric associativity maps, induction gives

$$\|u_1 \otimes \cdots \otimes u_m\|_{\mathcal{I}} \le K^{m-1} \prod_{j=1}^{m} \|u_j\|_{\mathcal{I}},$$

and membership in $\mathcal{I}$ at each step. Hence $\mathcal{I}$ is tensorstable. □

Corollary 1. *For a p-normed operator ideal, s-tensorstability and tensorstability are equivalent. Consequently condition (c) of [1, Theorem 7.8] may equivalently be replaced by tensorstability.*

Proof. Theorem 1 proves the previously open implication. The converse is [1, Proposition 7.12]. □

5 Verification, scope, and novelty

The proof is noncomputational. Its load-bearing points are the p-triangle estimate (2), the two uniform mixed-component maps of Lemma 1, and the exact intertwining identity (4). The accompanying `verification.md` checks these steps separately, including conventions, scalar fields, completions, and iteration. The constant is intentionally safe and is not asserted to be optimal.

Bounded novelty check. On 11 August 2026, the local run indexes and the downloaded arXiv source corpus were searched by arXiv id, title, authors, and the exact phrases "s-tensorstable", "symmetrically tensorstable", and the verbatim open question. The terminology occurred only in the source paper in the local corpus. External exact-phrase and title/author searches likewise found the source but no separate answer. OpenAlex metadata listed nine works citing the journal article; their titles and available records did not identify an answer. A directly relevant open 2020 dissertation on associativity of projective and injective tensor products was downloaded and text-searched and contained none of the exact terminology. No prior statement of Theorem 1 or of this block-diagonal mixed-component proof was found. Novelty confidence is moderate rather than definitive; specialist review of non-indexed literature is recommended.

Human-review recommendation. Check the averaged convention for $\vee$ and hence the factor 2 in Q; check that the source's norm-one map ι_Y^2 has the stated direction; and check whether later operator-ideal literature records this equivalence under different terminology. No unresolved mathematical dependency is known.

References

[1] G. Botelho and E. R. Torres, *Two-sided polynomial ideals on Banach spaces*, J. Math. Anal. Appl. **462** (2018), 900–914, doi:10.1016/j.jmaa.2017.12.054; circulated as *Polynomial ideals from a nonlinear viewpoint*, arXiv:1603.00246.